1. Core Regulatory Position (SAHPRA & Medicines Act)

a. GLP-1 Products Must Be Registered or Properly Compounded

✔ GLP-1 agonists like *semaglutide* and *tirzepatide* are **S4 prescription medicines** in South Africa — meaning they are **prescription-only** and must be dispensed via licensed professionals.

✔ **Compounding Allowed Only Under Section 14(4):**
Compounded medicines must:

- Relate to a *specific prescription* for a *specific patient* (max ~30 days supply)
- Only contain the **same active ingredient** as in a SAHPRA-registered medicine.

❌ SAHPRA has expressed intent to classify compounded GLP-1 products as *undesirable* under Section 23, meaning broad compounding for resale may soon be prohibited entirely.

📌 **ILLEGAL PRACTICE:** Compounding and dispensing *unregistered active ingredients*, or large-scale compounding for general supply without valid prescriptions, is likely illegal under South African legislation.

✅ 2. Operational Stock Structure: Meaningful Legal Framework

You want a multi-tenant/internal structure like a *simulated pharmacy chain* — but must remain compliant with SAPC/SAHPRA. Here's a *compliant architecture*:

a. Physical Warehouse ("Compounding/Dispensing Hub")

This is where your GLP-1 inventory lives.

✔ Licensed dispensing entity (e.g., a registered pharmacy or pharmacy-licensed operating unit) is responsible for:

- Stock control
- Purchase orders

- Cold chain management
- Licensing compliance

Legal Note: GLP-1 vials must arrive (or be compounded) with appropriate SAHPRA documentation and cold-chain validation.

---

b. Internal Sub-Warehouses (Simulated 'Pharmacy Branches' / Doctor Pools)

Instead of creating an *actual pharmacy branch network* with dispensing rights (which require SAPC licenses), you treat these as **internal "virtual pharmacies"** for operational modeling only.

Role Definitions:

Entity	Function	Legal Status
Pharmacy Warehouse (physically licensed)	Actual stock holding & dispensing point	Fully regulated
Internal Clinic / Doctor Unit	Can *request* stock for patients	*Not* a dispensing point
Doctor "Dispense Pools"	Authorization quotas tied to prescriptions	Operational only

Doctors **do not dispense stock directly to patients** — unless they are operating as licensed dispensers. Instead they **request stock which pharmacy dispenses against a prescription**.

✔ Physical dispensing always originates from the **Pharmacy Warehouse**.
✔ Internal units/tenants *request*, *track*, and *issue patient-specific orders*, but *stock never leaves system without pharmacist involvement*.
✔ Capturing stock movements (ERPNext) makes auditing and SAHPRA reporting straightforward.

✅ 3. ERPNext / Healthcare App Model – Module Design

a. Stock & Accounting

- **Warehouse Doc Types:**
  • Pharmacy Central Warehouse (licensed)
  • Virtual Sub-Warehouses (Clinics/Doctor Pools)
  • Patient Pick-Up Bins

- **Stock Movement Rules:**
  • Doctors create *Prescription Orders* (linked to patient).
  • These flow to **Pharmacy Pick List**.
  • Pharmacy **issues** stock → patient receives at pick-up or clinical consult.

✔ This ensures *stock never leaves without pharmacist sign-off*.

---

b. Multi-Tenant Logic

In ERPNext (or Frappe):

- **Tenant Entity:**
  • Clinics
  • Doctors
  • Pharmacy Units

- **Permissions:**
  • Doctors: *Can create prescription requisitions*
  • Pharmacists: *Can approve and output stock*
  • QA/Compliance: *Can audit dispensing logs*

This *simulation of a chain* is purely operational — **dispensing remains in regulated pharmacy context**, protecting you legally.

✅ 4. Process Flow: Compliant Dispensing

1️⃣ **Doctor writes prescription in system →**
2️⃣ **Doctor submits patient prescription order →**
3️⃣ **Pharmacy reviews & validates prescription →**
4️⃣ **Pharmacist dispenses GLP-1 stock →**
5️⃣ **Dispensed stock out of central warehouse only →**
6️⃣ **Patient receives medication + counseling**

Each event must be recorded with:

- Prescription ID
- Dispensed quantity
- Batch & expiry
- Pharmacist license
- Patient record

This is *required pharmacy practice*.

✅ 5. Legal & Compliance Considerations

Compounding & Quality

✔ Compounding requires strict records, sterility validation, and must **only use the registered active form** of semaglutide/tirzepatide.

⚠ SAHPRA has flagged risks with substandard and falsified GLP-1 products — strict quality control and supplier validation are essential.

🧠 Summary: Legal-Compliant System Design

Requirement	Your ERP/Stock Architecture
Maintain compliance with SAHPRA Medicines Act	Yes — by centralizing dispensing at licensed pharmacy
Support doctor direct patient dispensing	Yes — doctors *order*, but certified pharmacists dispense
Simulate multi-tenant pharmacy chain for operations	Yes — virtual warehouses, internal permissions
Track prescriptions & cold-chain stock	Yes — stock, batch, audit logs
Avoid illegal supply of compounded GLP-1	Yes — by enforcing prescription & licensed dispensing at central point

---

🛠️ Suggested ERPNext Doc Types

DocType	Purpose
Patient Prescription	Doctor enters script info
Clinic Requisition	Internal request to pharmacy
Pharmacy Pick List	Pharmacist selects stock to dispense
Stock Ledger Entry	Tracks all movements & audits
Compliance Audit Log	Tracks SAHPRA required reporting
Cold Chain Check Log	Temperature & handling audit

I'll break this into **6 layers**:

1. Regulatory grounding (concise, factual)
2. Legal operating model (what is *actually allowed*)
3. Physical vs logical stock architecture
4. ERPNext / Frappe system architecture
5. End-to-end dispensing workflow
6. **Compliance checkpoints (this is the key part)**

---

1. Regulatory Grounding (South Africa – Non-Negotiables)

1.1 Medicine classification

- GLP-1 agonists (Semaglutide, Tirzepatide)
  → **Schedule 4 medicines**

- Governed by:
    - Medicines and Related Substances Act, 1965
    - SAHPRA regulations
    - SAPC (South African Pharmacy Council) Good Pharmacy Practice (GPP)

1.2 Compounding reality (critical)

Compounding is **not a supply chain activity**. It is legally:

- Patient-specific
- Prescription-linked
- Quantity-limited (usually ≤30 days)
- Done by or under supervision of a **licensed pharmacist**

⚠️ **Key constraint**
A compounding facility **cannot legally behave like a wholesaler** supplying multiple downstream entities unless:

- Each dispense maps to a **named patient + prescription**
- Dispensing happens under a **licensed pharmacy entity**

This single fact drives your entire architecture.

---

2. Legal Operating Model (What You Are Actually Doing)

You are **not**:

- Running a pharmacy chain

- Running a wholesaler
- Allowing doctors to freely dispense medicine stock

You **are**:

- Operating **one licensed pharmacy / dispensing entity**
- Serving multiple clinics/doctors **operationally**
- Using software to *simulate* branches for control, reporting, and scale

> **Important framing**
> The system must always be able to answer:
> "Which licensed pharmacy dispensed this medicine to which patient, under which prescription, supervised by which pharmacist?"

If the answer is ever unclear → you are exposed.

---

3. Physical vs Logical Stock Architecture

This distinction is the heart of your design.

3.1 Physical Reality (Legal Truth)

There is **only ONE place where stock legally exists**:

✅ Central Licensed Pharmacy / Dispensing Facility

- Holds physical GLP-1 stock
- Controls cold chain
- Performs compounding (if applicable)
- Dispenses to patients

All stock ledger truth originates here.

3.2 Logical Reality (ERP / Multi-Tenant Simulation)

Everything else is **logical**, not legal:

Layer	Purpose	Legal Meaning
Virtual Pharmacy Branch	Operational grouping	NOT a licensed pharmacy
Clinic / Doctor Pool	Demand signal	NOT stock ownership
Patient Allocation	Reservation only	NOT physical stock

Doctors never "own" stock
They only **trigger demand**.

---

4. ERPNext / Frappe Architecture (Core Design)

4.1 Warehouse Model (Very Important)

Warehouses

1. **PHARM-CENTRAL-COLD**
    - Type: Physical, Licensed
    - Stock Ledger: REAL
    - Batch, Expiry, Temperature mandatory
2. **VIRTUAL-CLINIC-XXX**
    - Type: Logical

    - Stock Ledger: DISABLED or “Shadow”
    - No physical transfer allowed
3. **PATIENT-ALLOCATION (Virtual)**
    - Reservation bucket only
    - Never increases physical stock

📌 **Engineering rule**

Only one warehouse can ever reduce actual stock quantities.

4.2 Item Master (GLP-1)

Each GLP-1 item must enforce:

- Batch tracking = Mandatory
- Expiry date = Mandatory
- Serial = Optional (Batch is usually enough)
- Cold chain flag = TRUE
- Schedule = S4

4.3 Actors & Roles (Strict Separation)

Role	Allowed Actions
Doctor	Create prescription, request dispense
Pharmacist	Validate, compound, dispense
Clinic Admin	View status only

Promoter / Sales	**NO ACCESS** to prescriptions or stock
Compliance Officer	Read-only full audit

---

5. End-to-End Dispensing Workflow (System Truth)

Below is the **canonical workflow** your system must enforce.

STEP 1: Clinical Decision (Doctor)

DocType: `Patient Prescription`

Required fields:

- Patient (ID)
- Diagnosis
- Medicine (GLP-1)
- Dosage
- Duration
- Doctor registration number
- Doctor role validation

✅ Checkpoint 1 — Prescription Validity

System must validate:

- Doctor is licensed
- Medicine is S4

- Quantity does not exceed allowed compounding window

❌ If any fail → cannot proceed

STEP 2: Dispense Request (Doctor → Pharmacy)

DocType: `Dispense Request`

This is **NOT stock movement**.

Fields:

- Prescription reference
- Requested quantity
- Preferred clinic / delivery point
- Urgency

✅ Checkpoint 2 — No Stock Ownership

System rule:

> Dispense Request must NOT reserve or reduce stock

It is a *signal*, not an action.

STEP 3: Pharmacist Review

DocType: `Pharmacy Review`

Actions:

- Verify prescription
- Check patient history

- Confirm dosing appropriateness
- Decide:
    - Dispense
    - Reject
    - Request clarification

✅ Checkpoint 3 — Pharmacist Authority

System must enforce:

- Only users with Pharmacist role can approve
- Pharmacist license number logged immutably

STEP 4: Compounding (If Applicable)

DocType: `Compounding Record`

Fields:

- Source batch
- Compound date/time
- Compound quantity
- Beyond-use date (BUD)
- Responsible pharmacist

✅ Checkpoint 4 — Compounding Legality

System validation:

- Compounding must reference:
    - Prescription
    - Patient
- Quantity ≤ prescription limit
- Active ingredient matches registered product

STEP 5: Dispensing (THE ONLY REAL STOCK MOVE)

DocType: `Stock Entry (Dispense)`

Movement:

```
PHARM-CENTRAL-COLD  →  CONSUMED (Patient)
```

Not:

```
PHARM → CLINIC → PATIENT ❌
```

Mandatory fields:

- Prescription ID
- Patient ID
- Pharmacist ID
- Batch
- Expiry
- Quantity

✅ Checkpoint 5 — Single Source of Truth

System rule:

> All GLP-1 stock ledger reductions must originate from PHARM-CENTRAL-COLD

STEP 6: Patient Handover

DocType: `Dispense Confirmation`

Includes:

- Patient acknowledgment
- Counseling record
- Storage instructions
- Adverse event notes (if any)

✅ Checkpoint 6 — Proof of Dispense

No dispense is “complete” without:

- Confirmation record
- Patient linked
- Pharmacist signature

---

6. Compliance Checkpoints (System-Enforced)

This is what protects you in audits.

🔒 GLOBAL CHECKPOINTS (Always On)

CP-A: Prescription Lock

Once dispensing occurs:

- Prescription becomes read-only
- Quantity cannot be edited

CP-B: Batch Traceability

For every vial:

System must answer in <30 seconds:

- Where did it come from?
- Who compounded it?
- Who dispensed it?
- To which patient?

CP-C: Cold Chain Enforcement

- Temperature logs linked to batch
- Dispensing blocked if:
    - Temperature excursions unresolved

CP-D: Role Isolation

- Doctors: cannot see stock quantities

- Sales/promoters: cannot see prescriptions
- Clinics: cannot trigger stock movement

CP-E: Anti-Wholesaling Guard

Hard rule:

> A dispense record must ALWAYS reference a patient

No patient → no stock movement.

CP-F: SAHPRA Audit Mode

One click generates:

- Dispensed quantities per period
- Patient counts
- Pharmacist involvement
- Compounded vs pre-packed split

---

7. Why This Model Works (Strategically)

✔ Fully compliant with Medicines Act & GPP
✔ Scales like a pharmacy chain **without legally being one**
✔ Doctors feel like they “dispense” — but legally do not
✔ Stock risk and liability stays centralized
✔ Future-proof against SAHPRA tightening GLP-1 rules

1. **The golden rule of automation in regulated healthcare**
2. **End-to-end automated workflow (step-by-step)**

3. **Which parts can be auto vs must be human-approved**
4. **Exact system checkpoints & state transitions**
5. **How to implement this cleanly in Frappe (Workflows, Background Jobs, Guards)**

No fluff — this is production-grade thinking.

---

1. The Golden Rule (Non-Negotiable)

> **You may automate document creation, but never automate authority.**

That means:

- Documents can be *auto-generated*
- Transitions can be *queued*
- But **legal authority points** must:
  - Require a human
  - Capture license identity
  - Be immutable once executed

Your system should behave like an **aircraft autopilot**:

- 90% automated
- Critical actions require explicit human confirmation

---

2. High-Level Automated Flow (Bird's-Eye View)

```
Patient Signup
  ↓
Patient Intake Form (auto)
  ↓
Nurse Review (human gate)
  ↓
Suggested Prescription (draft)
  ↓
Doctor Review & Encounter (human gate)
  ↓
Prescription Approved (legal moment)
  ↓
Quote Generated (auto)
  ↓
Patient Acceptance (human gate)
  ↓
Sales Order → Delivery Note → Invoice (auto)
  ↓
Pharmacy Dispense Queue (auto)
  ↓
Pharmacist Approval & Dispense (human gate)
  ↓
Stock Ledger Updated (auto)
```

---

3. Detailed Workflow (With Automation Boundaries)

STEP 1: Patient Signup & Intake (Fully Automated)

DocType: `Patient`

- Created automatically on signup

DocType: `Patient Intake Form`

- Auto-generated
- Submitted by patient
- Locked after submission

✅ Checkpoint A — Data Integrity

System validations:

- Mandatory medical questions completed
- Consent accepted
- Identity verified (ID / DOB)

❌ No clinical processing until complete

STEP 2: Nurse Review (Human Gate #1)

DocType: `Intake Review`

Status:

- `Pending`
- `Approved`
- `Declined`
- `Needs More Info`

Actions:

- Nurse reviews intake
- Adds notes

- Can **draft** a Suggested Prescription

Important:

Nurse **cannot prescribe** — only *recommend*.

✅ Checkpoint B — Scope of Practice

System rule:

- Suggested Prescription = `Draft Only`
- No quantities finalized
- No pricing triggered

STEP 3: Doctor Encounter & Prescription (Human Gate #2)

DocTypes:

- `Patient Encounter`
- `Prescription`

Doctor actions:

- Review intake
- Review nurse suggestion
- Modify dosage, duration, medication
- Submit Encounter

🔐 LEGAL EVENT

When doctor submits the prescription:

- Doctor license is logged
- Timestamp locked
- Prescription becomes immutable

✅ Checkpoint C — Legal Authority Lock

System enforces:

- Only Doctor role can submit
- Prescription cannot be edited post-submit
- Quantity capped to allowed period

STEP 4: Automated Quote Generation (Auto)

Triggered by:

```
Prescription.status == Approved
```

DocType: `Quotation`

Auto-calculated:

- Medication cost
- Dispensing fee
- Delivery fee
- VAT (if applicable)

Rules:

- Quote references Prescription ID
- Quote validity window enforced

✅ Checkpoint D — Financial Traceability

Quote must reference:

- Prescription
- Patient
- Doctor

No orphan quotes allowed.

STEP 5: Patient Acceptance (Human Gate #3)

Patient actions:

- View quote
- Accept / Decline
- Payment initiation (if applicable)

Acceptance triggers:

- Sales Order creation (auto)

❌ No acceptance → no medical processing

STEP 6: Automated Sales Chain (Safe to Automate)

Once quote is accepted:

Auto-generated:

1. `Sales Order`
2. `Delivery Note`
3. `Packing Slip`
4. `Sales Invoice`

BUT:

These documents are **commercial**, not medical.

✅ Checkpoint E — Commercial vs Medical Separation

System rule:

- Financial docs cannot trigger stock movement
- They only *prepare* the dispense

STEP 7: Virtual Warehouse Allocation (Logical Only)

DocType: `Dispense Allocation`

System:

- Allocates required quantity
- Marks as “Reserved”
- No physical stock movement

Warehouse:

- `VIRTUAL-PHARMACY-QUEUE`

✅ Checkpoint F — Anti-Wholesaling Guard

Rule:

- Virtual warehouses cannot reduce stock
- Only allocation records allowed

STEP 8: Pharmacy Dispense Queue (Auto → Human)

DocType: `Pharmacy Dispense Task`

Auto-created when:

- Invoice exists
- Prescription exists
- Payment rules satisfied

Pharmacist sees:

- Patient
- Prescription
- Allocation
- Batch availability

STEP 9: Pharmacist Approval & Dispense (Human Gate #4)

Pharmacist actions:

- Validate prescription
- Select batch
- Confirm cold chain
- Approve dispense

🔐 LEGAL EVENT

This is the **only point where stock moves**.

DocType: `Stock Entry (Dispense)`

Movement:

`PHARM-CENTRAL-COLD → Patient (Consumed)`

✅ Checkpoint G — Single Source of Truth

System enforces:

- Warehouse must be licensed
- Batch mandatory
- Patient mandatory
- Pharmacist mandatory

STEP 10: Completion & Audit Trail (Auto)

Auto updates:

- Prescription → Fulfilled

- Sales Invoice → Completed
- Patient Medication History → Updated
- Compliance Log → Written

Immutable audit chain preserved.

---

4. What Is Automated vs Human-Controlled (Summary Table)

Step	Automated	Human
Intake form	✅	
Nurse review		✅
Suggested script		✅
Prescription approval		✅ (Doctor)
Quote generation	✅	
Quote acceptance		✅ (Patient)
Sales docs	✅	
Dispense queue	✅	
Stock dispense		✅ (Pharmacist)

---

5. How to Implement This in Frappe (Practically)

5.1 Use Workflow States, Not Custom Flags

Each critical DocType must have:

- Clear states
- Role-based transitions

Example (Prescription):

`Draft → Doctor Approved → Dispensed → Closed`

5.2 Use Server-Side Guards

Never rely on UI alone.

Examples:

- `validate()`: prevent illegal edits
- `before_submit()`: enforce role & license checks
- `on_submit()`: trigger safe automations
- `on_cancel()`: block medical reversals

5.3 Use Background Jobs for Automation

- Quote generation
- Sales chain creation
- Dispense task creation

This keeps UX fast and auditable.

5.4 Use Immutable Logs

For:

- Prescription approvals
- Dispense confirmations
- Pharmacist actions

Append-only logs only.

---

6. Why This Will Pass Audits

An auditor can see:

- Who decided clinically
- Who approved medically
- Who dispensed physically
- Who paid commercially
- And when each happened

No step crosses scope of practice.
No automation performs a regulated act.

---

two views:

1. **Master System State Machine (end-to-end)**
2. **Critical Sub-State Machines** (where legal authority lives)

I'll also annotate **LEGAL GATES** vs **AUTOMATION** so it's unambiguous.

1️⃣ Master End-to-End State Machine (System View)

This shows the **entire patient → dispense lifecycle**.

```
stateDiagram-v2
    [*] --> Patient_Onboarded

    Patient_Onboarded --> Intake_Submitted : Patient submits intake
    Intake_Submitted --> Intake_Under_Review : Auto-assign nurse

    Intake_Under_Review --> Intake_Declined : Nurse declines
    Intake_Under_Review --> Intake_Approved : Nurse approves

    Intake_Approved --> Suggested_Prescription_Drafted : Nurse
suggests (draft only)

    Suggested_Prescription_Drafted --> Doctor_Review

    Doctor_Review --> Prescription_Rejected
    Doctor_Review --> Prescription_Approved : Doctor submits encounter
⚠️ LEGAL

    Prescription_Approved --> Quote_Generated : Auto
    Quote_Generated --> Quote_Expired
    Quote_Generated --> Quote_Accepted : Patient accepts ⚠️ HUMAN

    Quote_Accepted --> Sales_Order_Created : Auto
    Sales_Order_Created --> Invoice_Issued : Auto

    Invoice_Issued --> Dispense_Queued : Auto (virtual allocation)

    Dispense_Queued --> Pharmacist_Review
    Pharmacist_Review --> Dispense_Rejected
    Pharmacist_Review --> Dispensed : Pharmacist approves ⚠️ LEGAL

    Dispensed --> Completed
    Completed --> [*]
```

2️⃣ Intake Review Sub-State Machine (Nurse Scope)

Purpose: Gate clinical suitability
Legal principle: Nurses cannot prescribe

```
stateDiagram-v2
    [*] --> Pending_Review
    Pending_Review --> Needs_More_Info
    Needs_More_Info --> Pending_Review
    Pending_Review --> Declined
    Pending_Review --> Approved

    Approved --> Suggested_Prescription_Draft
```

🔒 Compliance Rules

- Suggested Prescription = **Draft**
- No quantities finalized
- No pricing
- No stock impact

3️⃣ Prescription Authority State Machine (Doctor Scope)

This is a legal decision point

```
stateDiagram-v2
    [*] --> Draft
    Draft --> Under_Doctor_Review
    Under_Doctor_Review --> Rejected
    Under_Doctor_Review --> Approved : Doctor submit ⚠️ LEGAL
```

```
    Approved --> Locked
```

🔐 Compliance Locks

Once **Approved**:

- Prescription becomes immutable
- Doctor license ID captured
- Timestamp frozen
- Quantity capped to regulation

4️⃣ Commercial Automation State Machine (Safe to Automate)

Commercial ≠ Medical

```
stateDiagram-v2
    [*] --> Quote_Created
    Quote_Created --> Quote_Accepted
    Quote_Created --> Quote_Expired

    Quote_Accepted --> Sales_Order
    Sales_Order --> Delivery_Note
    Delivery_Note --> Invoice
```

✅ Automation Allowed

- No authority exercised
- No medicine dispensed
- No clinical judgement

5️⃣ Dispensing State Machine (Pharmacy Scope)

Second and final legal authority point

```
stateDiagram-v2
    [*] --> Dispense_Queued

    Dispense_Queued --> Pharmacist_Review
    Pharmacist_Review --> Clarification_Requested
    Clarification_Requested --> Pharmacist_Review

    Pharmacist_Review --> Dispense_Rejected
    Pharmacist_Review --> Dispense_Approved : Pharmacist confirm ⚠️
LEGAL

    Dispense_Approved --> Stock_Moved
    Stock_Moved --> Dispense_Completed
```

🔒 Hard Rules

- Only **licensed pharmacist**
- Mandatory batch + expiry
- Cold chain validation
- Patient reference required

6️⃣ Stock Ledger State Machine (Single Source of Truth)

```
stateDiagram-v2
    [*] --> Available

    Available --> Reserved : Virtual allocation only
    Reserved --> Available : Allocation cancelled
```

```
Reserved --> Dispensed : Pharmacist approval
Dispensed --> [*]
```

🚫 Illegal Transitions (Blocked by System)

- Available → Clinic
- Reserved → Clinic
- Clinic → Patient

Only:

`PHARM-CENTRAL-COLD → Patient (Consumed)`

7️⃣ Legal Authority Overlay (Very Important)

State Transition	Authority	Automated?
Intake approval	Nurse	❌
Prescription approval	Doctor	❌
Quote acceptance	Patient	❌
Dispense approval	Pharmacist	❌
Everything else	System	✅

8️⃣ Why This State Machine Is Strong

✔ Every **regulated act** is a state transition
✔ Every **regulated state** requires a licensed role
✔ Automation never crosses scope of practice
✔ Auditors can replay the entire journey deterministically

production-grade outputs, not hand-wavy examples.

I'll deliver this in **4 sections**, exactly as requested:

1. **Frappe Workflow JSON (key DocTypes)**
2. **DocType → State Machine mapping**
3. **Permission matrices per state**
4. **Audit replay report design (auditor-ready)**

Everything below is **legally aligned** with what we designed earlier.

1️⃣ Frappe Workflow JSON (Canonical Examples)

Below are **representative, correct Frappe workflow definitions**.
You would register these via **Workflow** DocType or fixtures.

1.1 Patient Intake Form – Workflow JSON

```
{
  "workflow_name": "Patient Intake Review",
  "document_type": "Patient Intake Form",
  "is_active": 1,
  "workflow_state_field": "workflow_state",
  "states": [
    {
      "state": "Submitted",
      "style": "Primary",
      "doc_status": 0,
      "allow_edit": "Patient"
    },
    {
```

```
    "state": "Under Nurse Review",
    "style": "Warning",
    "doc_status": 0,
    "allow_edit": "Nurse"
  },
  {
    "state": "Approved",
    "style": "Success",
    "doc_status": 1,
    "allow_edit": "Nurse"
  },
  {
    "state": "Declined",
    "style": "Danger",
    "doc_status": 1,
    "allow_edit": "Nurse"
  }
],
"transitions": [
  {
    "state": "Submitted",
    "action": "Assign Nurse",
    "next_state": "Under Nurse Review",
    "allowed": "System"
  },
  {
    "state": "Under Nurse Review",
    "action": "Approve",
    "next_state": "Approved",
    "allowed": "Nurse"
  },
  {
    "state": "Under Nurse Review",
    "action": "Decline",
    "next_state": "Declined",
    "allowed": "Nurse"
  }
]
```

```
}
```

🔒 Server-side guards (mandatory)

- Prevent edits after `Approved / Declined`
- Validate nurse role + registration

1.2 Prescription – Workflow JSON (LEGAL CORE)

```
{
  "workflow_name": "Prescription Approval",
  "document_type": "Prescription",
  "is_active": 1,
  "workflow_state_field": "workflow_state",
  "states": [
    {
      "state": "Draft",
      "style": "Warning",
      "doc_status": 0,
      "allow_edit": "Doctor"
    },
    {
      "state": "Doctor Approved",
      "style": "Success",
      "doc_status": 1,
      "allow_edit": ""
    },
    {
      "state": "Rejected",
      "style": "Danger",
      "doc_status": 1,
      "allow_edit": "Doctor"
    },
    {
      "state": "Dispensed",
```

```
      "style": "Primary",
      "doc_status": 1,
      "allow_edit": ""
    }
  ],
  "transitions": [
    {
      "state": "Draft",
      "action": "Approve Prescription",
      "next_state": "Doctor Approved",
      "allowed": "Doctor"
    },
    {
      "state": "Draft",
      "action": "Reject",
      "next_state": "Rejected",
      "allowed": "Doctor"
    },
    {
      "state": "Doctor Approved",
      "action": "Mark Dispensed",
      "next_state": "Dispensed",
      "allowed": "Pharmacist"
    }
  ]
}
```

🔐 Legal enforcement (code)

- `before_submit`: verify doctor license
- `on_submit`: lock quantity, dosage, medication
- No cancel allowed once approved

1.3 Pharmacy Dispense Task – Workflow JSON

```
{
  "workflow_name": "Pharmacy Dispense",
  "document_type": "Pharmacy Dispense Task",
  "is_active": 1,
  "workflow_state_field": "workflow_state",
  "states": [
    {
      "state": "Queued",
      "style": "Warning",
      "doc_status": 0,
      "allow_edit": "Pharmacist"
    },
    {
      "state": "Under Review",
      "style": "Primary",
      "doc_status": 0,
      "allow_edit": "Pharmacist"
    },
    {
      "state": "Dispensed",
      "style": "Success",
      "doc_status": 1,
      "allow_edit": ""
    },
    {
      "state": "Rejected",
      "style": "Danger",
      "doc_status": 1,
      "allow_edit": "Pharmacist"
    }
  ],
  "transitions": [
    {
      "state": "Queued",
      "action": "Start Review",
      "next_state": "Under Review",
      "allowed": "Pharmacist"
    },
```

```
    {
      "state": "Under Review",
      "action": "Dispense",
      "next_state": "Dispensed",
      "allowed": "Pharmacist"
    },
    {
      "state": "Under Review",
      "action": "Reject",
      "next_state": "Rejected",
      "allowed": "Pharmacist"
    }
  ]
}
```

2️⃣ DocType → State Machine Mapping

DocType	State Machine Purpose
Patient Intake Form	Clinical suitability gate
Intake Review	Nurse authority
Prescription	Doctor legal authority
Quotation	Patient consent
Sales Order	Commercial commitment
Delivery Note	Logistics
Sales Invoice	Financial record
Dispense Allocation	Virtual reservation
Pharmacy Dispense Task	Pharmacist authority
Stock Entry (Dispense)	Physical stock movement

Compliance Audit Log	Immutable evidence

---

3 Permission Matrices (Per State)

This is **where most systems fail** — yours won't.

3.1 Prescription Permissions

State	Doctor	Nurse	Pharmacist	Admin
Draft	Read/Write	Read	Read	Read
Doctor Approved	Read	Read	Read	Read
Dispensed	Read	Read	Read	Read

❌ No one can edit after approval.

3.2 Pharmacy Dispense Task Permissions

State	Pharmacist	Doctor	Admin
Queued	Read	Read	Read
Under Review	Read/Write	Read	Read
Dispensed	Read	Read	Read

3.3 Stock Entry (Dispense)

Role	Permission
Pharmacist	Create / Submit

Admin	Read
Doctor	❌
Nurse	❌
Sales	❌

4️⃣ Audit Replay Report Design (This Is Gold)

This is what **wins audits**.

4.1 Audit Replay Report: “Prescription → Dispense Trace”

Input Filters

- Patient
- Date range
- Medication
- Pharmacist
- Doctor

Output Sections (Chronological)

Section 1: Patient Onboarding

- Patient ID
- Intake submission timestamp
- Nurse reviewer + decision

Section 2: Clinical Authority

- Encounter ID
- Prescription ID
- Doctor name + license
- Approved timestamp
- Dosage + duration

Section 3: Patient Consent

- Quote ID
- Quote acceptance timestamp
- Payment confirmation

Section 4: Dispensing Authority

- Dispense Task ID
- Pharmacist name + license
- Batch number
- Expiry
- Cold chain status

Section 5: Stock Ledger Proof

- Stock Entry ID
- Source warehouse
- Quantity reduced
- Ledger timestamp

Key Feature: Replay Mode

Click “Replay” → system reconstructs the entire flow step-by-step

This answers:

- Who decided?
- Who approved?
- Who dispensed?
- Was it legal?

5️⃣ Engineering Non-Negotiables (Final Notes)

- **Never** auto-submit Prescription or Dispense docs
- **Never** allow cancellation after medical approval
- **Never** move stock from virtual warehouses
- **Always** log license numbers at submit time
- **Always** use server-side validation